

Lóránt Nagy, PhD

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Mathematician with a strong command of probability, stochastic modelling, mathematical finance and modern statistical and machine-learning methods, combining analytical thinking with practical expertise in developing models and operationalizing machine learning workflows.

PhD – Mathematical Finance 2018–2022
Central European University

MSc – Applied Mathematics 2015–2018
Budapest University of Technology and Economics

Professional Experience

Eötvös Loránd University, ELTE AI Research Group 2023–Present
Research Assistant

- Research deep-learning methods for statistical inference in stochastic-process and long-memory time-series models.
- Conducted an interpretability study on neural networks applied to time-series data: investigated which characteristics of the input drive the networks' inference.
- Led the technical development of reinforcement-learning methods for automated trading in stylized market environments, driving the project from problem formulation through model development to experimental validation.

Alfréd Rényi Institute of Mathematics 2019–Present
Research Fellow

- Investigate alternative stochastic dynamics for diffusion-based generative models.
- Conduct research in mathematical finance, stochastic control, market frictions, and optimal investment.

Artificial Intelligence National Laboratory 2024
ML Researcher

- Collaborated with ALTEO on intraday electricity-price forecasting.
- Developed and evaluated machine-learning regression models, with a focus on predictive reliability.

Sysler Kft. 2023
Research Consultant

- Prototyped a Bayesian late-fusion framework combining outputs from independent medical-diagnostic models to infer disease progression from empirical patient trajectories.

Morgan Stanley 2017
Interest Rate Modelling Intern

- Reviewed financial-product documentation, checking model assumptions, calculations, pricing results, and internal consistency.
- Reverse-engineered parts of a legacy pricing system written in A+ to understand existing model and pricing logic.

Technical Skills

- Expertise in statistical **analysis**, statistical learning, **probability** theory, stochastic **optimization**, mathematical finance, information theory
- Experience in machine learning operations, working knowledge of CI/CD, version control, containerizing applications, automated training, evaluation, and experiment tracking, hyperparameter optimization (Docker, Azure Machine Learning, TensorBoard, Optuna, MLflow, PyTorch, GitHub Actions)
- Strong understanding of the *mathematical* foundations of classical and modern **machine learning methods**: reinforcement learning, ensemble models, late fusion, Bayesian filtering, variational inference, diffusion generative models